E3 Solutions: geometry, proof and conic sections

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Computational answers were checked with SymPy.

Practice

1. By Thales, $\angle ACB = 90°$, so $AB = \sqrt{36+64} = 10$ and the radius is 5. $\tan\angle CAB = \frac{BC}{AC} = \frac{8}{6}$, so $\angle CAB \approx 53.1°$.

2. (a) $\frac{1}{2}\cdot 110° = 55°$.

(b) Then $\angle ACB$ is an inscribed angle on the major arc, whose central angle is $360° - 110° = 250°$: $\angle ACB = 125°$. (Equivalently: C on the minor arc and any point on the major arc form a cyclic quadrilateral with A and B, so the angles add to $180°$.)

3. Opposite angles add to $180°$: $\angle C = 180° - 78° = 102°$, $\angle D = 180° - 95° = 85°$. (Check: total $360°$.)

4. By the tangent–chord theorem $\angle ACB = 40°$. For D on the other arc, $ACBD$ is a cyclic quadrilateral, so $\angle ADB = 180° - 40° = 140°$.

5. $\sqrt{x^2+(y-2)^2} = |y+2| \Rightarrow x^2 - 4y + 4 = 4y + 4 \Rightarrow x^2 = 8y$, i.e. $y = \frac{1}{8}x^2$. For $y = \frac{1}{4}x^2$: $x^2 = 4y = 4py$ with $p = 1$: focus $(0,1)$, directrix $y = -1$.

6. $a = 5$, $b = 3$, $c = \sqrt{25-9} = 4$: foci $(\pm 4, 0)$, and $PF_1 + PF_2 = 2a = 10$.

7. $a = 4$, $b = 3$, $c = \sqrt{16+9} = 5$: foci $(\pm 5, 0)$; asymptotes $y = \pm\frac{3}{4}x$.

8. $(2\cdot 6 - 3\cdot 5,\ 3\cdot 4 - 1\cdot 6,\ 1\cdot 5 - 2\cdot 4) = (-3, 6, -3)$. Checks: $(-3,6,-3)\cdot(1,2,3) = -3+12-9 = 0$ and $(-3,6,-3)\cdot(4,5,6) = -12 + 30 - 18 = 0$. ✓

9. $\overrightarrow{AB} = (-1,2,0)$, $\overrightarrow{AC} = (-1,0,3)$, $\vec{n} = \overrightarrow{AB}\times\overrightarrow{AC} = (2\cdot 3 - 0\cdot 0,\ 0\cdot(-1) - (-1)\cdot 3,\ (-1)\cdot 0 - 2\cdot(-1)) = (6,3,2)$. Plane: $6x + 3y + 2z = d$, with A: $d = 6$. So $6x + 3y + 2z = 6$; B: 6 ✓, C: 6 ✓.

10. (a) $d = \dfrac{|2 - 1 + 2 - 9|}{\sqrt{4+1+4}} = \dfrac{6}{3} = 2$.

(b) Substitute: $2(1+\lambda) - (1+2\lambda) + 2(1+2\lambda) = 3 + 4\lambda = 9$, so $\lambda = \frac{3}{2}$ and the point is $(\frac{5}{2}, 4, 4)$. Check: $5 - 4 + 8 = 9$ ✓. Angle: $\sin\theta = \dfrac{|(1,2,2)\cdot(2,-1,2)|}{3\cdot 3} = \dfrac{4}{9}$, so $\theta \approx 26.4°$.

Exam-level

11. $\overrightarrow{CA} = -\vec{a} - \vec{c}$ and $\overrightarrow{CB} = \vec{a} - \vec{c}$. Then

$$\overrightarrow{CA}\cdot\overrightarrow{CB} = (-\vec{a}-\vec{c})\cdot(\vec{a}-\vec{c}) = -|\vec{a}|^2 + \vec{a}\cdot\vec{c} - \vec{c}\cdot\vec{a} + |\vec{c}|^2 = |\vec{c}|^2 - |\vec{a}|^2.$$

For $C \neq A, B$: $\angle ACB = 90° \iff \overrightarrow{CA}\cdot\overrightarrow{CB} = 0 \iff |\vec{c}| = |\vec{a}| \iff MC = MA$, i.e. C is on the circle with diameter AB. This proves Thales and its converse at once.

12. *Gegeven:* acute triangle ABC, $AD \perp BC$, $BE \perp AC$, $H = AD \cap BE$.

(a) $\angle ADB = 90°$ and $\angle AEB = 90°$ (altitudes). By the converse of Thales, D and E both lie on the circle with diameter AB. So A, B, D, E are concyclic.

(b) In quadrilateral $CDHE$: $\angle HDC = 90°$ and $\angle HEC = 90°$, so opposite angles add up to $180°$. By the converse of the cyclic quadrilateral theorem, C, D, H, E are concyclic. (Or: both D and E lie on the Thales circle of CH.)

(c) $ABDE$ is a cyclic quadrilateral (part (a); since the triangle is acute, D is between B and C and E between A and C, so the vertices are in the order A, B, D, E around the circle). So $\angle EAB + \angle BDE = 180°$. Since B, D, C are collinear, $\angle CDE = 180° - \angle BDE = \angle EAB = \angle CAB$. Together with the common angle at C, triangles CDE and CAB are similar (hh).

13. *Gegeven:* circles Γ_1, Γ_2 meeting in P, Q; $A \in \Gamma_1$, $B \in \Gamma_2$ on a line through P, P between A and B. In Γ_1, $\angle QAP$ is an inscribed angle on the fixed chord QP, so it is the same for all positions of A on the same arc (inscribed angle theorem). Likewise $\angle QBP$ is constant in Γ_2. Since P lies on segment AB, $\angle QAB = \angle QAP$ and $\angle QBA = \angle QBP$. In triangle AQB:

$$\angle AQB = 180° - \angle QAP - \angle QBP,$$

which is constant. (In fact all triangles AQB obtained this way are similar, and Q is the centre of the spiral similarity taking Γ_1 to Γ_2; compare E2 §10.)

14. Let $\vec{a}, \vec{b}, \vec{c}, \vec{h}$ be position vectors and H the intersection of the altitudes from A and B, so

$$(\vec{h} - \vec{a}) \cdot (\vec{b} - \vec{c}) = 0, \qquad (\vec{h} - \vec{b}) \cdot (\vec{c} - \vec{a}) = 0.$$

Add the two equations and expand:

$$\vec{h}\cdot\vec{b} - \vec{h}\cdot\vec{c} - \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} + \vec{h}\cdot\vec{c} - \vec{h}\cdot\vec{a} - \vec{b}\cdot\vec{c} + \vec{b}\cdot\vec{a} = 0,$$

which simplifies to $\vec{h}\cdot(\vec{b}-\vec{a}) - \vec{c}\cdot(\vec{b}-\vec{a}) = 0$, i.e. $(\vec{h}-\vec{c})\cdot(\vec{b}-\vec{a}) = 0$. So $CH \perp AB$ (or $H = C$, in a right-angled triangle): H lies on the third altitude too.

15. Let the quadrilateral be $ABCD$ with position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$, and midpoints $P = \frac{1}{2}(\vec{a}+\vec{b})$, $Q = \frac{1}{2}(\vec{b}+\vec{c})$, $R = \frac{1}{2}(\vec{c}+\vec{d})$, $S = \frac{1}{2}(\vec{d}+\vec{a})$. Then

$$\overrightarrow{PQ} = \tfrac{1}{2}(\vec{c}-\vec{a}) = \overrightarrow{SR}.$$

Equal vectors: PQ and SR are parallel and equally long, so $PQRS$ is a parallelogram. (Both sides are parallel to the diagonal AC and half as long.) The proof works for non-convex and even non-planar quadrilaterals.

16. $x^2 + y^2 = 4((x-3)^2 + y^2) \iff 3x^2 + 3y^2 - 24x + 36 = 0 \iff x^2 - 8x + y^2 + 12 = 0 \iff (x-4)^2 + y^2 = 4$. All points of this circle satisfy the original condition (the squaring was between non-negative quantities). The locus is the circle with centre $(4, 0)$ and radius 2 (circle of Apollonius). Check: $(2,0)$: $PA = 2$, $PB = 1$ ✓; $(6,0)$: $6 = 2 \cdot 3$ ✓.

17. A circle with centre (x, y) touching the x-axis has radius $|y|$; passing through F means $\sqrt{x^2 + (y-2)^2} = |y|$. Squaring: $x^2 - 4y + 4 = 0$, i.e. $y = \frac{1}{4}x^2 + 1$. A parabola: the centre is equidistant from the point $F(0,2)$ and the line $y = 0$, so the focus is $F(0,2)$ and the directrix is the x-axis; the vertex is $(0,1)$.

18. (a) Implicit differentiation: $2yy' = 4$, so at $(4,4)$: $y' = \frac{1}{2}$. Tangent: $y = \frac{1}{2}x + 2$.

(b) $y = 0$ gives $T(-4, 0)$. $FT = 1 - (-4) = 5$ and $FP = \sqrt{3^2 + 4^2} = 5$. ✓

(c) Triangle FPT is isosceles with $FT = FP$, so $\angle FPT = \angle FTP$. The horizontal line h through P is parallel to the x-axis, so the angle between h and the tangent equals $\angle FTP$ (corresponding angles, *F-hoeken*). Hence the tangent makes equal angles with PF and with h: a ray along h is reflected through F.

19. (a) $c = 3$, and $(0, 4)$ is an endpoint of the minor axis, so $b = 4$, $a^2 = b^2 + c^2 = 25$: $\dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$.

(b) $\frac{9}{25} + \frac{256/25}{16} = \frac{9}{25} + \frac{16}{25} = 1$ ✓. Tangent: $\dfrac{3x}{25} + \dfrac{(16/5)y}{16} = 1$, i.e. $\dfrac{3x}{25} + \dfrac{y}{5} = 1$ (or $3x + 5y = 25$).

(c) $PF_2 = \frac{16}{5}$ (vertical distance, $F_2 = (3, 0)$); $PF_1 = \sqrt{36 + \frac{256}{25}} = \sqrt{\frac{1156}{25}} = \frac{34}{5}$. Sum: $\frac{50}{5} = 10 = 2a$ ✓.

20. (a) $\frac{25}{9} - \frac{256/9}{16} = \frac{25-16}{9} = 1$ ✓. $c^2 = 9 + 16 = 25$: $F_{1,2} = (\mp 5, 0)$. $PF_2 = \frac{16}{3}$, $PF_1 = \sqrt{100 + \frac{256}{9}} = \frac{34}{3}$; difference $\frac{18}{3} = 6 = 2a$ ✓.

(b) Tangent: $\dfrac{5x}{9} - \dfrac{(16/3)y}{16} = 1$, i.e. $\dfrac{5x}{9} - \dfrac{y}{3} = 1$. With $y = 0$: $T(\frac{9}{5}, 0)$.

(c) $TF_1 = \frac{9}{5} + 5 = \frac{34}{5}$, $TF_2 = 5 - \frac{9}{5} = \frac{16}{5}$, so $TF_1 : TF_2 = 34 : 16 = PF_1 : PF_2$. T lies between F_1 and F_2. The internal bisector of $\angle F_1PF_2$ meets F_1F_2 in the unique point dividing it internally in this ratio, which is T. So line PT, the tangent, is the internal bisector.

21. Normals $(1, 1, 0)$ and $(0, 1, 1)$: $\cos\theta = \dfrac{1}{\sqrt{2}\sqrt{2}} = \dfrac{1}{2}$, $\theta = 60°$.

22. Direction: $(1, 1, 1) \times (1, -1, 2) = (1 \cdot 2 - 1 \cdot (-1),\ 1 \cdot 1 - 1 \cdot 2,\ 1 \cdot (-1) - 1 \cdot 1) = (3, -1, -2)$. A point: put $z = 0$: $x + y = 6$, $x - y = 3$, so $x = \frac{9}{2}$, $y = \frac{3}{2}$. Line: $\vec{x} = (\frac{9}{2}, \frac{3}{2}, 0) + \lambda(3, -1, -2)$. Check: direction is perpendicular to both normals ($3 - 1 - 2 = 0$, $3 + 1 - 4 = 0$) ✓.

23. (a) $\overrightarrow{AB} = (-1, 1, 0)$, $\overrightarrow{AC} = (-1, 0, 1)$, cross product $(1, 1, 1)$ with length $\sqrt{3}$. Area $= \frac{\sqrt{3}}{2}$.

(b) $\frac{1}{6}|((1, 0, 0) \times (0, 1, 0)) \cdot (0, 0, 1)| = \frac{1}{6}$.

(c) Volume $= \frac{1}{3} \cdot \text{area} \cdot h$: $\frac{1}{6} = \frac{1}{3} \cdot \frac{\sqrt{3}}{2} h$, so $h = \frac{1}{\sqrt{3}}$. The plane is $x + y + z = 1$; distance formula: $\frac{|0-1|}{\sqrt{3}} = \frac{1}{\sqrt{3}}$ ✓.

Challenge

24. Let t be the external bisector of $\angle F_1PF_2$ at P, i.e. the line through P bisecting the angle between ray PF_2 and the ray PX opposite to PF_1. Let F_2' be the reflection of F_2 in t.

- Reflection in t swaps the two rays that t bisects, so it maps ray PF_2 onto ray PX. Hence F_2' lies on ray PX, i.e. on line F_1P beyond P, with $PF_2' = PF_2$.
- Therefore $F_1F_2' = F_1P + PF_2' = PF_1 + PF_2 = 2a$.
- Let $Q \neq P$ be on t. Since t is the perpendicular bisector of F_2F_2' (or $F_2 = F_2'$ lies on t, which happens only in the degenerate case P on line F_1F_2), $QF_2 = QF_2'$. Then

$$QF_1 + QF_2 = QF_1 + QF_2' \geq F_1F_2' = 2a,$$

with equality only if Q is on segment F_1F_2'. But that segment meets t only at P (line F_1F_2' is not t, since $F_1 \notin t$: t is perpendicular to the internal bisector at P). So $QF_1 + QF_2 > 2a$: Q lies outside the ellipse.

- So t meets the closed region $E = \{X : XF_1 + XF_2 \le 2a\}$ only in P. The region E is convex: if $X, Y \in E$ and $Z = (1-s)X + sY$ with $0 \le s \le 1$, then by the triangle inequality $ZF_1 \le (1-s)XF_1 + sYF_1$, similarly for F_2, and adding gives $ZF_1 + ZF_2 \le 2a$. A line through a boundary point of a convex region that does not enter the region is a supporting line, and at a point where the boundary is a smooth curve the only supporting line is the tangent.

So the tangent at P is the external bisector: the angle between the tangent and PF_1 equals the angle between the tangent and PF_2. Light from F_1 reflects to F_2.

25. Choose K on diagonal AC with $\angle ABK = \angle DBC$.

- *Triangles ABK and DBC:* $\angle ABK = \angle DBC$ (construction) and $\angle BAK = \angle BAC = \angle BDC$ (inscribed angles on arc BC). So they are similar (hh), and $\dfrac{AK}{AB} = \dfrac{DC}{DB}$, i.e. $AK \cdot BD = AB \cdot CD$.
- *Triangles KBC and ABD:* adding $\angle KBD$ to both sides of $\angle ABK = \angle DBC$ gives $\angle ABD = \angle KBC$; and $\angle BCK = \angle BCA = \angle BDA$ (inscribed angles on arc AB). So they are similar, and $\dfrac{CK}{CB} = \dfrac{DA}{DB}$, i.e. $CK \cdot BD = AD \cdot BC$.
- Add: $(AK + KC) \cdot BD = AB \cdot CD + AD \cdot BC$, and $AK + KC = AC$. ■

Rectangle with sides p, q and diagonal d: $d \cdot d = p \cdot p + q \cdot q$. Pythagoras.

26. (a) Write out both sides. Left:

$$(a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2.$$

Right: $(a_1^2+a_2^2+a_3^2)(b_1^2+b_2^2+b_3^2) - (a_1b_1+a_2b_2+a_3b_3)^2$. In the right side the terms $a_i^2b_i^2$ cancel, leaving $\sum_{i\neq j} a_i^2b_j^2 - 2\sum_{i<j} a_ib_ia_jb_j = \sum_{i<j}(a_i^2b_j^2 + a_j^2b_i^2 - 2a_ib_ja_jb_i) = \sum_{i<j}(a_ib_j - a_jb_i)^2$, which is the left side. (SymPy confirms the difference expands to 0.)

(b) With $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$: $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2(1 - \cos^2\theta) = |\vec{a}|^2|\vec{b}|^2 \sin^2\theta$, and $\sin\theta \ge 0$ for $0 \le \theta \le \pi$. The parallelogram has base $|\vec{a}|$ and height $|\vec{b}| \sin\theta$.

(c) The left side of (a) is ≥ 0, so $(\vec{a} \cdot \vec{b})^2 \le |\vec{a}|^2|\vec{b}|^2$. Equality iff $\vec{a} \times \vec{b} = \vec{0}$, i.e. iff $\vec{a}$ and $\vec{b}$ are parallel (one is a multiple of the other). Note that this proves Cauchy–Schwarz *without* using angles, which is good: it is what justifies defining the angle by $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$ in the first place.